

Bison

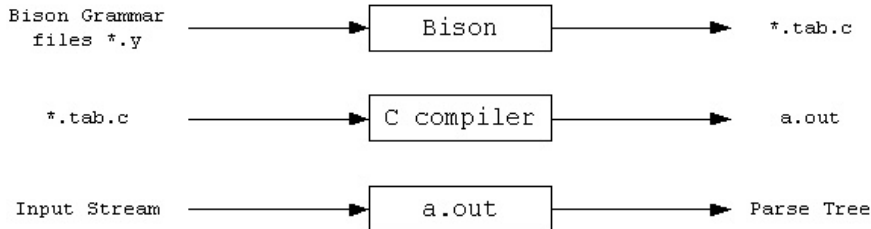
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Bison

- Bison is a parser generator
- Bison is a bottom-up parser
- Accepts LALR(1) grammars and emits a C program to parse that grammar



Format of the input file

```
%{  
C Declarations  
%}  
Bison Declarations  
%%  
Grammar Rules  
%%  
Additional C Code
```

- **Useful Bison definitions:** `%token`, `%union`, `%type`, `%left`, `%right`, `%nonassoc`, ...
- **Important data structures and functions:** `yylval`, `YYSTYPE`, `yyperror()`, `yyparse()`

Compiling the program

Assume that the flex file is `p1.l` and the bison file is `p1.y`. The compilation steps are:

```
$ flex p1.l # produces lex.yy.c, which is  
            # included in p1.y through #include  
$ bison p1.y # produces p1.tab.c  
$ gcc p1.tab.c -o p1.exe -lfl
```